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Problem Set 2
Macroeconomics I
Due before class, Wednesday, November 11, 2026

Recall the single-sector neoclassical model with search from class. In the first problem set, we solved this model using a linear approximation to the model around its non-stochastic steady state.

Because we log-linearized our model, the impulse response in PS1 part (d) is equivalent to a perfect foresight solution of the approximate linearized model. Now, we are going to solve the model equations exactly. To do this, we assume that agents have *perfect foresight* and use a shooting algorithm.

The parameters you should use in this problem set are listed below, they are the same as in Problem Set 1. As before, we target the steady-state unemployment rate $u = 1 - N^{ss}$ and back out the value of χ consistent with that target. Please follow the same instructions as in Problem Set 1 for reporting your answers to the problem set.

Table 1: Numerical Parameter Values

Concept	Symbol	Value
Discount factor	β	0.99
Inverse IES	σ	2.00
Capital share	α	0.30
Capital depreciation	δ_k	0.03
Labor separation	δ_n	0.10
Vacancy cost	ϕ_n	0.50
Steady-state unemployment	u	0.06
Matching function elasticity	ε	0.25
$\log A$ persistence	ρ	0.95
$\log A$ shock s.d.	σ_a	0.01

Note: in building your code, you are free to use and modify the example programs on the course website. *See the syllabus for the course policy on AI tools.*

1. **Shooting Method:**

(a) Generate a function called `residual`. The function should accept three input arguments. The first argument is `XYv`, a (vectorized) candidate path for all of the endogenous variables in $[X(t+1), Y(t)]$ for periods 1 through 500. The second argument is the vector steady-state values of `XYss`. The third is the object containing the model parameters, `param`.

The output of the function should be a residual vector, `resid`, that evaluates each model equation $F(\cdot)$ along the history, under the assumption that K_1 is at steady-state, $A_1 = exp(\sigma_a)$, and all the elements of $Y(501)$ are at steady-state.

The `numel(XYv)` should be the same as the `numel(resid)` of the return vector `resid`.

(b) Create a function handle to your `residual` function, which fixes arguments 2-3. Using your linearized solutions from PS1, generate a "guess" for the path of X_t and Y_t – the inputs to your `residual` function – in response to a 1.0% initial increase in TFP. Your loss function at this initial guess should evaluate to

$$\texttt{sum(abs(resid0(:)))} = 0.00X_1 62693X_2 279$$

To confirm your answer, **report** the two missing digits represented by X_i's above.

(c) Use the matlab command `fsolve` to find a path for the endogenous variables that (numerically exactly) satisfies the model equations at each point in time, based on this initial disturbance.

(d) **Generate and report** a 4x2 figure in Matlab. In each subplot, plot the impulse response of one the main model variables, $[[A, K, N], [Y, C, I, N, V]]$, according to the linearized approximate and exact perfect foresight model solution. Use a solid line for the former and a dashed line with $\times$'s for the later. In a sentence or two, **assess** the quality of the linear approximation for shocks of this size.

(e) Now, increase the size of the initial productivity shock from 1% to 10%, and **generate and report** the same figure as in 1(d). **Compare** the quality of the linearization for the smaller and larger shocks.

(f) *Can you find a parameterization that makes the non-linear effects noticeably larger, even when the shock size is 1%? **Describe** the key parameters you need to change, and their values. Which variable shows the largest non-linearity? **Provide** some intuition for why these parameters matter for the quality of the approximation.